



Prehospital Supraglottic Airways: An NAEMSP Position Statement and Resource Document

John W. Lyng, Kimberly T. Baldino, Darren Braude, Christie Fritz, Juan A. March, Timothy D. Peterson & Allen Yee

To cite this article: John W. Lyng, Kimberly T. Baldino, Darren Braude, Christie Fritz, Juan A. March, Timothy D. Peterson & Allen Yee (2022) Prehospital Supraglottic Airways: An NAEMSP Position Statement and Resource Document, Prehospital Emergency Care, 26:sup1, 32-41, DOI: [10.1080/10903127.2021.1983680](https://doi.org/10.1080/10903127.2021.1983680)

To link to this article: <https://doi.org/10.1080/10903127.2021.1983680>



© 2021 The Author(s). Published with license by Taylor & Francis Group, LLC.



Published online: 10 Jan 2022.



Submit your article to this journal [↗](#)



Article views: 10486



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 12 View citing articles [↗](#)

PREHOSPITAL SUPRAGLOTTIC AIRWAYS: AN NAEMSP POSITION STATEMENT AND RESOURCE DOCUMENT

John W. Lyng , Kimberly T. Baldino , Darren Braude, Christie Fritz ,
Juan A. March , Timothy D. Peterson, and Allen Yee

ABSTRACT

Supraglottic airway (SGA) devices provide effective conduits for oxygenation and ventilation and may offer protection from gastric aspiration. SGA devices are widely used by EMS clinicians as both rescue and primary airway management devices. While in common use for more than four decades, major developments in SGA education, science, and technology have influenced clinical strategies of SGA insertion and use in prehospital airway management for patients of all ages.

NAEMSP recommends:

- SGAs have utility as a primary or secondary EMS airway intervention. EMS agencies should select SGA strategies that best suit available resources and local clinician skillset, as well as the nature of their clinical practice setting.
- EMS agencies that perform endotracheal intubation must also equip their clinicians with SGA devices and ensure adequate training and competence.

- In select situations, drug-assisted airway management may be used by properly credentialed EMS clinicians to facilitate SGA insertion.
- Confirmation of initial and continuous SGA placement using waveform capnography is strongly encouraged as a best practice.
- When it is functioning properly, EMS clinicians should refrain from converting an SGA to an endotracheal tube. The decision to convert an SGA to an endotracheal tube must consider the patient's condition, the effectiveness of SGA ventilations, and the clinical context and course of initial SGA insertion
- SGA training, competency, and clinical use must be continuously evaluated by EMS agencies using focused quality management programs.

Key words: supraglottic airway; RSA; EMS; SGA; airway management

PREHOSPITAL EMERGENCY CARE 2022;26:32–41

Received August 10, 2021 from Department of Emergency Medicine, University of Minnesota School of Medicine, Minneapolis, MN, USA (JWL); National Association of EMS Physicians®, Overland Park, KS, USA (JWL, KTB, DB, CF, JAM, TDP, AY); St Joseph's University Medical Center, Hackensack Meridian School of Medicine, Rowan University School of Osteopathic Medicine, Patterson, NJ, USA (KTB); Departments of Emergency Medicine and Anesthesiology, University of New Mexico, Albuquerque, NM, USA (DB); Beth Israel Deaconess Medical Center EMS Division, Boston, MA, USA (CF); Division of EMS Department of Emergency Medicine Brody School of Medicine, East Carolina University, Greenville, NC, USA (JAM); Southeast Region EMS Council, Sitka, AK; Capital City Fire & Rescue, Juneau, AK; Assistant Clinical Professor, University of New Mexico School of Medicine, Albuquerque, NM, USA (TDP); Chesterfield Fire and EMS, Virginia Commonwealth University Health System, Richmond, VA, USA (AY). Revision received September 7, 2021; accepted for publication September 17, 2021.

Approved by: National Association of EMS Physicians – 20 July 2021

Address correspondence to John W. Lyng, MD, Paramedic, at jlyngmd@gmail.com

© 2021 The Author(s). Published with license by Taylor & Francis Group, LLC.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way.

doi:10.1080/10903127.2021.1983680

INTRODUCTION

Supraglottic airways (SGAs) are advanced airway devices intended to be inserted into the oropharynx without the use of laryngoscopy. While varying in design, SGAs are devised to situate in the hypopharyngeal space overlying or outside of the glottic opening, providing indirect oxygen delivery to the trachea. SGAs have also been referred to as “blind insertion,” “extraglottic,” “periglottic,” and “infraglottic” airway devices. While these terms are technically more precise, they are functionally synonymous, and “supraglottic airway” is the more widely used term. While traditionally used as a rescue device after failed intubation attempts, several factors have resulted in increasing use of SGAs in the prehospital setting (1–3).

UTILITY AND SELECTION OF SGAs

SGAs have utility as a primary or secondary EMS airway intervention. EMS agencies should select SGA strategies that best suit available resources and local clinician skillset, as well as the nature of their clinical practice setting.

The decision to use an SGA as a primary or secondary prehospital airway intervention depends on several factors including clinician skills and

opportunities to perform airway interventions. Although older SGAs like the Combitube were developed as primary airway devices, historically paramedics were typically instructed to prioritize use of ETI and only used SGAs as rescue devices. Several studies support use of SGAs as rescue airways in the setting of failed ETI (4, 5).

Over the past decade three factors have helped to shift use of SGAs from a role as rescue devices for failed ETI to a role as the primary airway intervention in the EMS setting: 1) newer, functionally superior devices (King LTS-D, i-gel, and various LMAs), 2) better recognition of the pitfalls of ETI in the EMS setting, and 3) several randomized controlled trials that demonstrated better outcomes and faster processes of care with SGA over ETI in out-of-hospital cardiac arrest (OHCA).

There is mixed evidence that using SGAs may be more easily and rapidly placed than ETTs (6, 7). In contrast to Frascione et al. who found no difference in success rate or time to insertion when comparing the King LTS-D to ETI, Russi et al. identified that placement of a Combitube is faster than ETI (8, 9). The ease and speed of SGA insertion is likely dependent on features of the device and the competency of the EMS clinician as shown by Russi et al. who showed that insertion of a Combitube is slower than that of a single lumen SGA and by March et al. who showed that i-gel insertion was faster than the King LT (8, 10). Studies by Jarvis et al., Bernhard et al., and Middleton et al. have also shown that the success of SGA placement, including both first-pass and overall success rates, varies between different SGA devices (11–13). Though these studies suggest that various device-specific factors may affect training needs and successful use, no studies have demonstrated the superiority of any particular SGA device.

Factors that may influence EMS agency use and selection of SGAs include the skillset and scope of practice of the clinicians (e.g., ALS vs BLS); whether the clinicians also perform ETI, including how often and how well they perform it; integration with neighboring first responder, rescue, and transport agencies; whether the clinicians can perform drug-assisted airway management; and whether the agency intends for SGAs to be used as a primary or secondary airway.

SGAS AND EMS CLINICIAN SCOPE OF PRACTICE

In the United States, the NHTSA 2019 *National EMS Scope of Practice Model* establishes insertion of SGAs as

one of the minimum competencies for Advanced EMT (AEMT) and Paramedic level clinicians. However, the guidelines exclude use of SGAs as a minimum competency from Emergency Medical Technician (EMT) and Emergency Medical Responder (EMR) practice (14). This decision was based on concerns regarding added time and expense of educating EMTs and EMRs in SGA use and in the technology needed to ensure appropriate SGA placement and function. However, the *National Scope of Practice Model* defines the minimum competencies, not the highest-permitted skillset for each clinician type. In fact, several studies have demonstrated successful use of SGAs by basic-level clinicians and use of SGAs by EMTs and EMRs might be reasonable if certain conditions are met (15–19).

Patients who might be appropriate for use of an SGA by a BLS clinician include OHCA patients and patients who are severely obtunded by severe head trauma or severe sedative-hypnotic toxicity. Other patients may require use of drug-assisted airway management by ALS clinicians to facilitate placement of an SGA. It must be recognized, though, that training and use of SGAs cannot substitute for training and competent use of bag-valve-mask ventilation by all types of EMS clinicians (15).

The decision to include or exclude SGA use ultimately falls upon more local levels of EMS governance. Several states, regional EMS authorities, and local EMS agencies have included SGAs within EMR and EMT scope of practice. Outside the U.S. SGAs are included as a minimum competency for basic level EMS clinicians though their use by such clinicians is predicated on use of end-tidal capnography to confirm device placement (20, 21).

AVAILABILITY OF SGA DEVICES

EMS agencies that perform endotracheal intubation must also equip their clinicians with SGA devices and ensure adequate training and competence.

EMS ETI success rates range from 53% to 90%, indicating that a substantial portion of cases will require management with rescue airway interventions (22–25). Though all EMS clinicians should be facile with manual ventilation using a bag-valve-mask device, there are circumstances where BVM use may prove difficult (26). Historically some EMS agencies have also used surgical airway techniques as a backup strategy for failed ETI; however, these are difficult to perform, even for the most experienced clinicians. Many studies have described

successful use of SGAs after failed ETI, and SGAs are recommended as rescue devices for failed ETI by several organizations (4, 22, 23, 27).

Use of SGAs should be protocolized and integrated with ETI strategies, such as in the algorithmic approach to airway management suggested by Wang et al. (28). Planning for failed intubation, including ensuring availability of both BVM ventilation and supraglottic airways, should always be part of the pre-intubation briefing. Further, a 2020 joint organizational position statement regarding essential equipment for ground BLS and ALS ambulances recommends that ALS ambulances should be stocked with supraglottic airways in sizes to fit neonates to adults (29).

DRUG ASSISTED SGA INSERTION

In select situations, drug assisted airway management may be used by properly credentialed EMS clinicians to facilitate SGA insertion.

Drug-assisted airway management, which includes more commonly known rapid sequence intubation (RSI), also includes Rapid Sequence Airway (RSA), the concept of using sedative and paralytic medications to facilitate SGA insertion by ALS clinicians. Series reported by Frascione and Braude affirm the safety and effectiveness of RSA (30–33). The most recent and largest case series published by Braude et al. reported a success rate of 94% and aspiration rates equivalent to other methods of emergency airway management (33). The potential ease and speed of SGA insertion vs ETI may make it easier to achieve first-pass success with RSA than RSI and may help reduce hypoxemia and airway trauma associated with prolonged and repeated intubation attempts (34).

The potential disadvantages of RSA include more difficult ventilation via an SGA if high airway pressures are required, inability to access the airway reliably for suction, and perceptions that SGAs are less effective in protecting against aspiration than endotracheal tubes. The risk of aspiration with SGA is lower than previously thought. Several studies suggest that if aspiration occurs it likely happens before invasive airway placement rather than during or following insertion (35–41). Second-generation SGAs with gastric aspiration lumens allow for evacuation of gastric contents, further reducing the risk for aspiration. The need for an Emergency Department (ED) exchange of the SGA to an endotracheal tube has also been suggested as a disadvantage of RSA; however, the controlled exchange to an ETT in the ED setting is likely not a major risk to the patient.

Prior to embarking on RSA, EMS clinicians should assess for risk factors of unsuccessful SGA placement and ALS clinicians should plan for potential failed RSA attempt by preparing for manual ventilation, ETI, or surgical airway procedures if a can't intubate/can't ventilate situation develops (42). Though RSI and RSA are used in nearly identical patient populations and clinical environments, there are no published direct comparisons of RSA and RSI strategies in the EMS setting. More research comparing RSA with RSI and other EMS-based drug-assisted airway management strategies is necessary to further describe the utility of this approach in the EMS setting and to define its effects on patient outcomes.

CONFIRMATION OF SGA PLACEMENT

Confirmation of initial and continuous SGA placement using waveform capnography is strongly encouraged as a best practice.

Extensive data inform the requirement for independent confirmation of ETT placement using waveform capnography and other subjective and objective measures of successful placement (43–45). Despite this emphasis on ETT confirmation, and possibly because of the simplicity of most SGA insertion techniques, many programs have placed relatively little attention on confirmation of SGA placement. However, emerging case reports and series highlight the pitfalls of SGA placement, including airway malposition or misplacement (1, 46, 47). Though initial confirmation of device location is important SGA dislodgement after initial successful placement is possible. These considerations motivate the recommendation for SGA placement confirmation both immediately after insertion and on a continuing basis.

Common methods for confirming SGA placement include auscultation of breath sounds, visualization of chest rise, and observation for presence of exhaled condensation in the lumen of the SGA, though these techniques vary in reliability and availability. Waveform capnography represents the most reliable method to confirm initial and ongoing placement of advanced airways and is recommended as a best practice for SGAs. Vithalani et al. were the first to use capnography to objectively describe the incidence of unrecognized misplaced supraglottic airways in the EMS setting. They identified a two-fold problem when subjective means of placement confirmation were used in isolation by clinicians: failed recognition of misplaced SGAs in

35% of cases, and unnecessary removal of properly placed SGAs in 1.5% of cases. Use of capnography would have helped correctly identify SGA placement in both of these situations. Though this study's findings are limited to a single EMS agency and evaluation of a single SGA device, the study highlights the importance of using objective measurements to confirm initial and ongoing SGA placement (1).

An important issue with requiring capnography to confirm SGA placement is that though both ALS and BLS clinicians may use SGAs, typically only ALS clinicians are trained in ETCO_2 interpretation and use. BLS clinicians who are inserting SGAs should ideally use waveform capnography for initial and ongoing airway placement confirmation, however lack of resources to train BLS clinicians to interpret waveform capnography as well as lack of funding to purchase waveform capnography equipment are barriers to adoption of this practice. In cases where waveform capnography is not included in local EMR or EMT scope of care, we recommend that BLS clinicians cautiously use alternative methods to assess SGA placement.

While recommended by some experts, colorimetric end-tidal carbon dioxide detection is unreliable for confirming SGA placement. First, it is possible to qualitatively detect enough residual carbon dioxide through an SGA to result in a "positive" colorimetric result even when the position of the device is suboptimal. Second, colorimetric EtCO_2 devices can give false-positive results in some settings and are not reliable if they become contaminated with liquids (48–50). Therefore, qualitative colorimetric EtCO_2 detection should not be used in isolation to assess the proper position and function of an SGA on either an initial or ongoing basis. The use of devices that provide continuous quantitative capnometry may help avoid these pitfalls of colorimetric devices, though there are no focused studies on use of such technology by BLS clinicians.

Although pulse oximetry is typically accessible to both BLS and ALS clinicians, determination of SGA placement should be based on assessment of adequate ventilation, not oxygen saturation. Poor oxygenation in the setting of adequate ventilation through an SGA usually indicates a failed oxygen supply or respiratory pathology resulting in a shunt, V/Q mismatch, or disordered diffusion, not dysfunction of the SGA itself. Additionally, pulse oximetry is limited by a significant lag-time between onset of inadequate ventilation and drops in pulse oximetry values (51–54). Also, failure to recognize that an SGA may be properly placed and providing

appropriate ventilation despite low pulse oximetry values may lead to premature removal of a functioning invasive airway (55).

Ideally proper SGA placement is determined by any EMS clinician by using waveform capnography in addition to assessment for effective chest rise, presence of symmetric lungs sounds, and absent signs of major air leakage (56).

CONVERSION OF AN SGA TO AN ETT

When it is functioning properly, EMS clinicians should refrain from converting an SGA to an endotracheal tube. The decision to convert an SGA to an ET tube must consider the patient's condition, the effectiveness of SGA ventilations, and the clinical context and course of initial SGA insertion.

If functioning properly, prehospital SGAs should not be exchanged for an ET tube in the prehospital setting. If an SGA is not properly ventilating and the function of the SGA cannot be improved via immediate troubleshooting the EMS clinician should remove the SGA and begin manual ventilation using a bag-valve-mask. In very rare circumstances exchange of a SGA for an ETT may be considered.

Clinical scenarios that might favor exchange include the need for high ventilation pressures, the need for tracheal suctioning, presence of copious oropharyngeal secretions, bleeding, or emesis, and anticipated swelling below the level of the SGA that might eventually render it ineffective for ventilation. While SGAs have been associated with significant airway swelling after prolonged placement, development of this complication typically requires several hours and prolonged transport is rarely an indication for SGA exchange. Complications related to duration of SGA placement have been reported by Gaither et al. for the King LT at 3 hours and by Gerstein et al. for the LMA-Fastrach at 5 hours. Braude et al. reported there were no complications associated with an LMA-Supreme SGA being left in place for 9 hours (57–59).

Any clinician considering SGA-ETT exchange should first determine why the SGA was originally placed. This is especially true if an SGA was placed as a rescue device after failed ETI attempts because if the patient underwent multiple unsuccessful intubation attempts, additional intubation attempts will likely be unsuccessful (23).

Regardless of the clinical setting, if an SGA-ETT exchange is determined to be necessary, clinicians

considering exchange should assess for potential difficulty in mask ventilation, intubation, or challenging surgical airway access in case exchange attempts fail (60–65). Further, the procedure should be tailored to the patient condition, clinician experience, available tools, and the specific SGA in situ. Inappropriately or prematurely performing an SGA-ETT exchange while a patient is hypoxic may result in further desaturation and anoxic injury or cardiac arrest. Prior to exchange clinicians should maximize patient oxygenation, decompress the stomach, consider use of drug-assisted airway management, and prepare contingency airway management plans.

Depending on the type of SGA being exchanged, several general approaches can be used including extraluminal, endoluminal, and bougie-assisted blind techniques. Exchange of an SGA via an extraluminal approach by displacing or removing the SGA and performing intubation via direct or video laryngoscopy may result in inadvertent loss of airway control. One endoluminal approach includes use of a fiberoptic bronchoscope and is best suited for exchanging most LMA-style SGAs, however this technology is not typically available in the prehospital setting (66, 67). Endoluminal bougie-assisted blind exchange has been described for the King LT, i-gel, and LMA-style SGAs, though use bougie-exchange of the King LT is discouraged due to significant risk for airway perforation (68–72). Blind direct passage of an endotracheal tube has been described for two devices, with success rates ranging from 15% to 90% (73–76). These techniques have been further well-described by Braude et al. and Driver et al. (56, 77).

SGA TRAINING AND CLINICAL USE

SGA training, competency, and clinical use must be continuously evaluated by EMS agencies using focused quality management programs.

Supraglottic airway use is a low-frequency, high risk intervention that should be subject to the same rigorous quality management practices that are used to oversee endotracheal intubation in the EMS setting, including use of high-quality training sessions and focused application of quality management programs to help reduce risk and improve patient outcomes. SGA quality management programs must ensure that clinicians receive appropriate training, that they are periodically assessed for their cognitive, psychomotor, and affective

competencies related to SGA use, and that clinical use is providing appropriate patient benefit while minimizing patient harms. The principles of prehospital airway management training and education and quality management of prehospital airway programs, including pediatric-focused topics, are discussed in companion documents to this manuscript, but certain SGA-specific concepts deserve further discussion.

Training and Competency. In order to safely and correctly utilize an SGA an EMS clinician must first demonstrate competency in several domains. These include the cognitive domain pertaining to recognition of SGA indications, contraindications, and complications; and the psychomotor domain pertaining to delivery of oxygen via nasal cannulas and face-masks, manual bag-valve-mask ventilation, oro/nasopharyngeal airway placement, and upper airway suctioning.

A number of studies have investigated the ability of EMS clinicians to perform SGA insertion, though many of these studies are limited to manikin-based training encounters (78–86). Notably, not all training manikins are alike, as some models are easier or harder for SGA insertion depending on the make/model of SGA being used (87). Further, successful SGA placement in a manikin may not translate to successful use on actual patients. A few studies have evaluated insertion of various SGA devices by different EMS clinicians on anesthetized patients in the operating room, however access to such clinical settings for purposes of EMS clinician training has become increasingly limited (12, 88, 89).

Training in the use of SGAs requires more than developing a psychomotor skill. EMS clinicians must also receive education in cognitive and critical thinking domains including knowledge of indications and contraindications for SGA use as defined in local protocol, recognition of and mitigation strategies for SGA malfunctions, and awareness of potential complications related to SGA use. Competency in less-invasive airway management strategies must also be demonstrated as entry-level proficiencies prior to embarking on SGA education and clinical use. Unfortunately published studies have typically focused only on the psychomotor component of SGA insertion in artificial environments and also have not assessed EMS clinician cognitive aptitudes pertaining to SGA use. This makes it difficult to extrapolate whether these studies support whether clinicians can perform SGA insertion in actual clinical practice. Further, attrition of cognitive knowledge and decay of psychomotor skill are significant issues that affect the successful use of

SGAs by EMS clinicians. However, studies by Fischer et al., Ruetzler et al., and Maddocks have shown that rates of skill decay may be affected by which type of SGA is being used and may be slower than decay of ETI skills (90–92).

Complications Associated with SGAs. In addition to establishing robust training and monitoring performance of EMS clinicians in successfully inserting SGAs, quality management programs should also maintain surveillance for potential complications of SGA use. A case series by Bernhard et al. describes several complications associated with King-style SGAs including airway obstruction due to inadvertent tracheal intubation, massive tongue and pharyngeal edema, air-leak associated hypoventilation, and device obstruction by foreign material (93). Other case reports of SGA-related injuries occurring in the prehospital and surgical settings include hypopharyngeal perforation, pneumomediastinum, subcutaneous emphysema, pneumoperitoneum, pneumothorax, upper airway bleeding, esophageal laceration or perforation, and pressure-related tissue injuries of the tongue, pharynx, and hypopharyngeal structures. (47, 94–98). Notably the Combitube has been shown to have up to a 40% complication rate (95–98).

One other important concern related to SGAs is the potential to impair cerebral blood flow. Two swine studies by Segal et al. and Kim et al. that measured carotid artery blood flow during induced cardiac arrest found that the King LTS-D, LMA, Combitube, and i-gel SGAs each significantly reduced carotid blood flow relative to the ETT (99, 100). However, several human studies evaluating the effect of SGAs on the cervical vasculature and blood flow have shown mixed results. Radiographic studies of SGAs by Niell et al. (MRI), and White et al. (CT scans) found no significant compression of the internal carotid artery. (101, 102). In several device-specific studies, Rasulo et al. (LMA-Unique) did not demonstrate significant reductions in cerebral blood flow, Eismann et al. (King LTS-D and LMA) showed no impairment of internal carotid artery blood flow, and Nandwani et al. (LMA) found no evidence that either the carotid artery or the internal jugular vein was compressed significantly (103–105). In contrast, Colbert et al. and Zhang et al. showed several different SGAs caused anatomic displacement of the cervical vessels and reduced the velocity of blood flow through them (106, 107). The variable findings of these studies suggest that the effect of SGAs on blood flow through the cervical vessels may be influenced by the design of the SGA (e.g., cuffed vs uncuffed) as

well as individual patient-specific factors. Unfortunately, this body of evidence only informs our understanding of the effect of SGAs on the cervical vessels and blood flow in hemodynamically stable patients, not on patients with low flow states such as shock or cardiac arrest. There is a significant need for prospective evaluation of cerebral perfusion in both hemodynamically stable and unstable patients whose airways are managed with SGAs.

Patient Outcomes. Medical directors should understand how SGA use may impact clinical outcomes of different patient populations and use this understanding when developing SGA-related protocols.

Cardiac Arrest. Studies by Abo, Wang, and others have shown significantly shorter interruptions in chest compressions during OHCA resuscitation when airways were managed with SGAs compared to ETTs (82, 108). Further, the Pragmatic Airway Resuscitation Trial (PART) by Wang et al. demonstrated that 72-hour survival was greater in patients receiving SGAs compared to ETTs (109). Secondary outcomes from PART also found increased rates of return of spontaneous circulation, hospital survival, and favorable neurologic exam with SGA management. However, a subsequent systematic review performed by Carney et al. showed no difference in outcomes when comparing bag-valve-mask manual ventilation, SGAs, and ETT use in OHCA patients (34). Yet another study by Bengert et al. also failed to show a survival benefit in OHCA when comparing use of the i-gel SGA to ETI (110). Finally, the AIRWAYS-2 trial by Bengert et al. did not identify a more favorable 30-day functional outcome for SGAs vs ETI in OHCA (110).

Trauma. Limited studies regarding use of SGAs in the trauma setting have shown potential associations of SGA use with worse patient outcomes, though it is unclear whether this is a reflection of the severity of patient injuries or if there is a more direct connection between outcomes and SGA use (111, 112).

FUTURE RESEARCH

Future research should attempt to identify which, if any, SGA devices are easiest to use by all levels of EMS clinician, and more importantly, whether any SGA device is superior to others with respect to positively affecting patient outcomes. Research is also needed to help identify techniques to assess SGA function and placement that are accessible to all levels of EMS clinician. Further, studies

investigating the best methods for exchanging SGAs for ETTs in the emergency department should also be conducted. Finally, there is a critical need for prospective research regarding the safety of SGAs in the EMS setting, especially their effect on cerebral blood flow in low-flow states such as shock and cardiac arrest.

CONCLUSION

In summary, SGAs can be used as a reasonable alternative to ETT by ALS clinicians and may expand the ability of BLS clinicians to perform better ventilation and oxygenation of certain patients. However, data regarding patient-based outcomes of SGAs compared to use of a bag-valve-mask or ETT remains mixed, and optimal SGA utilization strategies have yet to be fully defined.

ORCID

John W. Lyng  <http://orcid.org/0000-0001-5191-5700>

Kimberly T. Baldino  <http://orcid.org/0000-0003-3935-4538>

Christie Fritz  <http://orcid.org/0000-0002-0206-6164>

Juan A. March  <http://orcid.org/0000-0002-8780-4882>

References

- Vithalani VD, Vlk S, Davis SQ, Richmond NJ. Unrecognized failed airway management using a supraglottic airway device. *Resuscitation*. 2017;119:1–4. doi:10.1016/j.resuscitation.2017.07.019.
- Dunford JV, Davis DP, Ochs M, Doney M, Hoyt DB. Incidence of transient hypoxia and pulse rate reactivity during paramedic rapid sequence intubation. *Ann Emerg Med*. 2003;42(6):721–8. doi:10.1016/S0196-0644(03)00660-7.
- Deakin CD, King P, Thompson F. Prehospital advanced airway management by ambulance technicians and paramedics: is clinical practice sufficient to maintain skills? *Emerg Med J*. 2009;26(12):888–91. doi:10.1136/emj.2008.064642.
- Shavit I, Aviram E, Hoffmann Y, Biton O, Glassberg E. Laryngeal mask airway as a rescue device for failed endotracheal intubation during scene-to-hospital air transport of combat casualties. *Eur J Emerg Med*. 2018;25(5):368–71.
- Lockey D, Crewdson K, Weaver A, Davies G. Observational study of the success rates of intubation and failed intubation airway rescue techniques in 7256 attempted intubations of trauma patients by pre-hospital physicians. *Br J Anaesth*. 2014;113(2):220–5. doi:10.1093/bja/aeu227.
- Cook T, Howes B. Recent developments in efficacy and safety of supraglottic airway devices. *Contin Educ Anaesth Crit Care Pain*. 2011;11(2):56–61. doi:10.1093/bjaceaccp/mkq058.
- Wang HE, Szydlo D, Stouffer JA, Lin S, Carlson JN, Vaillancourt C, Sears G, Verbeek RP, Fowler R, Idris AH, ROC Investigators, et al. Endotracheal intubation versus supraglottic airway insertion in out-of-hospital cardiac arrest. *Resuscitation*. 2012;83(9):1061–6. doi:10.1016/j.resuscitation.2012.05.018.
- Russi CS, Miller L, Hartley MJ. A comparison of the King-LT to endotracheal intubation and combitube in a simulated difficult airway. *Prehosp Emerg Care*. 2008;12(1):35–41. doi:10.1080/10903120701710488.
- Frascone RJ, Russi C, Lick C, Conterato M, Wewerka SS, Griffith KR, Myers L, Connors J, Salzman JG. Comparison of prehospital insertion success rates and time to insertion between standard endotracheal intubation and a supraglottic airway. *Resuscitation*. 2011;82(12):1529–36. doi:10.1016/j.resuscitation.2011.07.009.
- March JA, Tassey TE, Resurreccion NB, Portela RC, Taylor SE. Comparison of the i-gel supraglottic and king Laryngotracheal airways in a simulated tactical environment. *Prehosp Emerg Care*. 2018;22(3):385–9. doi:10.1080/10903127.2017.1399183.
- Jarvis JL, Barton D, Wang H. Defining the plateau point: When are further attempts futile in out-of-hospital advanced airway management? *Resuscitation*. 2018;130:57–60. doi:10.1016/j.resuscitation.2018.07.002.
- Bernhard M, Gries A, Ramshorn-Zimmer A, Wenzel V, Hossfeld B. Insertion success of the laryngeal tube in emergency airway management. *Biomed Res Int*. 2016;2016:1–13. doi:10.1155/2016/3619159.
- Middleton PM, Simpson PM, Thomas RE, Bendall JC. Higher insertion success with the i-gel supraglottic airway in out-of-hospital cardiac arrest: a randomised controlled trial. *Resuscitation*. 2014;85(7):893–7. doi:10.1016/j.resuscitation.2014.02.021.
- National Association of State EMS Officials. National EMS Scope of Practice Model 2019 (Report No. DOT HS 812-666). Washington (DC): National Highway Traffic Safety Administration; 2019. https://www.ems.gov/pdf/National_EMS_Scope_of_Practice_Model_2019.pdf
- Fiala A, Lederer W, Neumayr A, Egger T, Neururer S, Toferer E, Baubin M, Paal P. EMT-led laryngeal tube vs. face-mask ventilation during cardiopulmonary resuscitation – a multicenter prospective randomized trial. *Scand J Trauma Resusc Emerg Med*. 2017;25(1):104. doi:10.1186/s13049-017-0446-1.
- Roth D, Hafner C, Aufmesser W, Hudabunigg K, Wutti C, Herkner H, Schreiber W. Safety and Feasibility of the laryngeal tube when used by EMTs during out-of-hospital cardiac arrest. *Am J Emerg Med*. 2015;33(8):1050–5. doi:10.1016/j.ajem.2015.04.048.
- Länkimäki S, Alahuhta S, Kurola J. Feasibility of a laryngeal tube for airway management during cardiac arrest by first responders. *Resuscitation*. 2013;84(4):446–9. doi:10.1016/j.resuscitation.2012.08.326.
- Gahan K, Studnek JR, Vandeventer S. King LT-D use by urban basic life support first responders as the primary airway device for out-of-hospital cardiac arrest. *Resuscitation*. 2011;82(12):1525–8. doi:10.1016/j.resuscitation.2011.06.036.
- Hart D, Driver B, Kartha G, Reardon R, Miner J. Efficacy of laryngeal tube versus bag mask ventilation by inexperienced providers. *WestJEM*. 2020;21(3):688–93. doi:10.5811/westjem.2020.3.45844.
- Service UKNH. East of England Ambulance Service: Scope of Practice Policy. Trust EoEASN, editor. National Health Service Trust, UK: East of England Ambulance Service: Scope of Practice Policy; 2021. <https://www.eastamb.nhs.uk/Policies/clinical/Scope-of-Practice-Policy.pdf>
- Association OP. Ontario Paramedics Scope of Practice. Ontario Paramedic Association, Toronto, ON. 2021. <https://>

- www.ontarioparamedic.ca/misc/Scope%20of%20Practice%20v4.1.pdf.
22. Ochs M, Davis D, Hoyt D, Bailey D, Marshall L, Rosen P. Paramedic-performed rapid sequence intubation of patients with severe head injuries. *Ann Emerg Med*. 2002;40(2):159–67. doi:10.1067/mem.2002.126397.
 23. Nwanne T, Jarvis J, Barton D, Donnelly JP, Wang HE. Advanced airway management success rates in a national cohort of emergency medical services agencies. *Resuscitation*. 2020;146:43–9. doi:10.1016/j.resuscitation.2019.11.006.
 24. Hubble MW, Brown L, Wilfong DA, Hertelendy A, Benner RW, Richards ME. A meta-analysis of prehospital airway control techniques Part I: orotracheal and nasotracheal intubation success rates. *Prehosp Emerg Care*. 2010;14(3):377–401. doi:10.3109/10903121003790173.
 25. Warner KJ, Sharar SR, Copass MK, Bulger EM. Prehospital management of the difficult airway: a prospective cohort study. *J Emerg Med*. 2009;36(3):257–65. doi:10.1016/j.jemermed.2007.10.058.
 26. Jabre P, Penaloza A, Pinero D, Duchateau F-X, Borron SW, Javaudin F, Richard O, de Longueville D, Bouilleau G, Devaud M-L, et al. Effect of bag-mask ventilation vs endotracheal intubation during cardiopulmonary resuscitation on neurological outcome after out-of-hospital cardiorespiratory arrest: a randomized clinical trial. *JAMA*. 2018;319(8):779–87. doi:10.1001/jama.2018.0156.
 27. Cooper RM. Preparation for and management of “failed” laryngoscopy and/or intubation. *Anesthesiology*. 2019;130(5):833–49. doi:10.1097/ALN.0000000000002555.
 28. Wang HE, Kupas DF, Greenwood MJ, Pinchall ME, Mullins T, Gluckman W, Sweeney TA, Hostler D. An algorithmic approach to prehospital airway management. *Prehosp Emerg Care*. 2005;9(2):145–55. doi:10.1080/10903120590924618.
 29. Lyng J, Adelgais K, Alter R, Beal J, Chung B, Gross T, et al. Recommended essential equipment for basic life support and advanced life support ground ambulances 2020: a joint position statement. *Prehosp Emerg Care*. 2021;25(3):451–459. doi:10.1080/10903127.2021.1886382.
 30. Braude D, Richards M. Rapid sequence airway (RSA)-a novel approach to prehospital airway management. *Prehosp Emerg Care*. 2007;11(2):250–2. doi:10.1080/10903120701206032.
 31. Frascone RJ, Wewerka SS, Burnett AM, Griffith KR, Salzman JG. Supraglottic airway device use as a primary airway during rapid sequence intubation. *Air Med J*. 2013;32(2):93–7. doi:10.1016/j.amj.2012.06.008.
 32. Frascone RJ, Wewerka SS, Griffith KR, Salzman JG. Use of the King LTS-D during medication-assisted airway management. *Prehosp Emerg Care*. 2009;13(4):541–5. doi:10.1080/10903120903144817.
 33. Braude D, Dixon D, Torres M, Martinez JP, O'Brien S, Bajema T. Brief research report: prehospital rapid sequence airway. *Prehosp Emerg Care*. 2021;25(4):583–7. doi:10.1080/10903127.2020.1792015.
 34. Carney N, Cheney T, Totten AM, Jungbauer R, Neth MR, Weeks C, et al. Prehospital airway management: a systematic review. *Prehosp Emerg Care*. 2021;1–28.
 35. Steuerwald MT, Braude DA, Petersen TR, Peterson K, Torres MA. Preliminary report: comparing aspiration rates between prehospital patients managed with extraglottic airway devices and endotracheal intubation. *Air Med J*. 2018;37(4):240–3. doi:10.1016/j.amj.2018.04.004.
 36. Steuerwald MT, Robinson BR, Hanseman DJ, Makley A, Pritts TA. Prehospital airway technique does not influence incidence of ventilator-associated pneumonia in trauma patients. *J Trauma Acute Care Surg*. 2016;80(2):283–8. doi:10.1097/TA.0000000000000886.
 37. Evans HL, Warner K, Bulger EM, Sharar SR, Maier RV, Cuschieri J. Pre-hospital intubation factors and pneumonia in trauma patients. *Surg Infect (Larchmt)*. 2011;12(5):339–44. doi:10.1089/sur.2010.074.
 38. Evans HL, Zonies DH, Warner KJ, Bulger EM, Sharar SR, Maier RV, et al. Timing of intubation and ventilator-associated pneumonia following injury. *Arch Surg*. 2010;145(11):1041–6. doi:10.1001/archsurg.2010.239.
 39. John RE, Hill S, Hughes TJ. Airway protection by the laryngeal mask. A barrier to dye placed in the pharynx. *Anaesthesia*. 1991;46(5):366–7. doi:10.1111/j.1365-2044.1991.tb09545.x.
 40. Mercer MH. An assessment of protection of the airway from aspiration of oropharyngeal contents using the Combitube airway. *Resuscitation*. 2001;51(2):135–8. doi:10.1016/s0300-9572(01)00390-2.
 41. Bercker S, Schmidbauer W, Volk T, Bogusch G, Bubser HP, Hensel M, Kerner T. A comparison of seal in seven supraglottic airway devices using a cadaver model of elevated esophageal pressure. *Anesth Analg*. 2008;106(2):445–8, table of contents. doi:10.1213/ane.0b013e3181602ae1.
 42. Martin-Gill C, Prunty HA, Ritter SC, Carlson JN, Guyette FX. Risk factors for unsuccessful prehospital laryngeal tube placement. *Resuscitation*. 2015;86:25–30. doi:10.1016/j.resuscitation.2014.10.015.
 43. Grmec S. Comparison of three different methods to confirm tracheal tube placement in emergency intubation. *Intensive Care Med*. 2002;28(6):701–4. doi:10.1007/s00134-002-1290-x.
 44. Li J. Capnography alone is imperfect for endotracheal tube placement confirmation during emergency intubation 11 Original Contributions is coordinated by John A. Marx, MD, of Carolinas Medical Center, Charlotte, North Carolina. *The Journal of Emergency Medicine*. 2001;20(3):223–9. doi:10.1016/S0736-4679(00)00318-8.
 45. DeBoer S, Seaver M, Arndt K. Verification of endotracheal tube placement: a comparison of confirmation techniques and devices. *J Emerg Nurs*. 2003;29(5):444–50. doi:10.1016/S0099-1767(03)00268-X.
 46. Van Zundert AA, Kumar CM, Van Zundert TC. Malpositioning of supraglottic airway devices: preventive and corrective strategies. *Br J Anaesth*. 2016;116(5):579–82. doi:10.1093/bja/aew104.
 47. Michalek P, Donaldson W, Vobrubova E, Hakl M. Complications associated with the use of supraglottic airway devices in perioperative medicine. *Biomed Res Int*. 2015;2015:746560. doi:10.1155/2015/746560.
 48. Keller WR, Biehler J, Linares MY-R, Garcia-Pena BM. False-positive colorimetric capnometry after ingestion of carbonated beverages. *Pediatr Emerg Care*. 2009;25(2):69–73.
 49. Hughes S, Blake B, Woods S, Lehmann C. False-positive results on colorimetric carbon dioxide analysis in neonatal resuscitation: potential for serious patient harm. *J Perinatol*. 2007;27(12):800–1. doi:10.1038/sj.jp.7211831.
 50. Vargese JH. Use of disposable end tidal carbon dioxide detector device for checking endotracheal tube placement. *J Clin Diagnos Res*. 2007;1:204–9.
 51. Sinex JE. Pulse oximetry: principles and limitations. *Am J Emerg Med*. 1999;17(1):59–67. doi:10.1016/s0735-6757(99)90019-0.
 52. Farmery AD, Roe PG. A model to describe the rate of oxyhaemoglobin desaturation during apnoea. *Br J Anaesth*. 1996;76(2):284–91. doi:10.1093/bja/76.2.284.

53. Cacho G, Perez-Calle JL, Barbado A, Lledo JL, Ojea R, Fernandez-Rodriguez CM. Capnography is superior to pulse oximetry for the detection of respiratory depression during colonoscopy. *Rev Esp Enferm Dig.* 2010;102(2):86–9.
54. Keidan I, Gravenstein D, Berkenstadt H, Ziv A, Shavit I, Sidi A. Supplemental oxygen compromises the use of pulse oximetry for detection of apnea and hypoventilation during sedation in simulated pediatric patients. *Pediatrics.* 2008;122(2):293–8. doi:10.1542/peds.2007-2385.
55. Aguilar SA, Davis DP. Latency of pulse oximetry signal with use of digital probes associated with inappropriate extubation during prehospital rapid sequence intubation in head injury patients: case examples. *J Emerg Med.* 2012;42(4):424–8. doi:10.1016/j.jemermed.2011.06.127.
56. Braude D, Steuerwald M, Wray T, Galgon R. Managing the out-of-hospital extraglottic airway device. *Ann Emerg Med.* 2019;74(3):416–22. doi:10.1016/j.annemergmed.2019.03.002.
57. Gaither JB, Matheson J, Eberhardt A, Colwell CB. Tongue engorgement associated with prolonged use of the King-LT laryngeal tube device. *Ann Emerg Med.* 2010;55(4):367–9. doi:10.1016/j.annemergmed.2009.07.009.
58. Gerstein NS, Braude D, Harding JS, Douglas A. Lingual ischemia from prolonged insertion of a fastrach laryngeal mask airway. *West J Emerg Med.* 2011;12(1):124–7.
59. Braude D, Southard A, Bajema T, Sims E, Martinez J. Rapid sequence airway using the LMA-supreme as a primary airway for 9 h in a multi-system trauma patient. *Resuscitation.* 2010;81(9):1217. doi:10.1016/j.resuscitation.2010.06.001.
60. Kheterpal S, Han R, Tremper KK, Shanks A, Tait AR, O'Reilly M, Ludwig TA. Incidence and predictors of difficult and impossible mask ventilation. *Anesthesiology.* 2006;105(5):885–91. doi:10.1097/0000542-200611000-00007.
61. Corso RM, Cattano D, Buccioli M, Carretta E, Maitan S. [Post analysis simulated correlation of the El-Ganzouri airway difficulty score with difficult airway]. *Rev Bras Anesthesiol.* 2016;66(3):298–303. doi:10.1016/j.bjan.2016.02.007.
62. el-Ganzouri AR, McCarthy RJ, Tuman KJ, Tanck EN, Ivankovich AD. Preoperative airway assessment: predictive value of a multivariate risk index. *Anesth Analg.* 1996;82(6):1197–204. doi:10.1097/0000539-199606000-00017.
63. El-Orbany M, Woehlck HJ. Difficult mask ventilation. *Anesth Analg.* 2009;109(6):1870–80. doi:10.1213/ANE.0b013e3181b5881c.
64. Vannucci A, Cavallone LF. Bedside predictors of difficult intubation: a systematic review. *Minerva Anesthesiol.* 2016;82(1):69–83.
65. Gupta S, Sharma R, Jain D. Airway assessment: predictors of difficult airway. *Indian J Anaesth.* 2005;49(4):257–62.
66. Ueki R, Komazawa N, Nishimoto K, Sugi T, Hirose M, Kaminoh Y. Utility of the Aintree Intubation Catheter in fiberoptic tracheal intubation through the three types of intubating supraglottic airways: a manikin simulation study. *J Anesth.* 2014;28(3):363–7. doi:10.1007/s00540-013-1724-3.
67. Koumpan Y, Murdoch J, Beyea JA, Kahn M, Colbeck J. A case report: establishing a definitive airway in a trauma patient with a king laryngeal tube in situ in the presence of a closed head injury and difficult airway: "Between the Devil and the Deep Blue Sea." *A&A Practice.* 2017;8(6):139–41.
68. Lopez NT, McCoy SK, Carroll C, Jones E, Miller JA. Non-conventional utilization of the aintree intubating catheter to facilitate exchange between three supraglottic airways and an endotracheal tube: a cadaveric trial. *Mil Med.* 2019;184(1-2):e222–e8. doi:10.1093/milmed/usy144.
69. Lutes M, Worman DJ. An unanticipated complication of a novel approach to airway management. *J Emerg Med.* 2010;38(2):222–4. doi:10.1016/j.jemermed.2008.08.011.
70. Wong DT, Yang JJ, Mak HY, Jagannathan N. Use of intubation introducers through a supraglottic airway to facilitate tracheal intubation: a brief review. *Can J Anaesth.* 2012;59(7):704–15. doi:10.1007/s12630-012-9714-8.
71. Howath A, Brimacombe J, Keller C. Gum-elastic bougie-guided insertion of the ProSeal laryngeal mask airway: a new technique. *Anaesth Intensive Care.* 2002;30(5):624–7. doi:10.1177/0310057X0203000514.
72. Stancil S, Miller J, Riddle M. Laryngeal mask airway exchange using a gum elastic bougie with a rotational twist technique. *J Spec Oper Med.* 2014;14(3):74–7.
73. Gerstein NS, Braude DA, Hung O, Sanders JC, Murphy MF. The fastrach intubating laryngeal mask airway: an overview and update. *Can J Anaesth.* 2010;57(6):588–601. doi:10.1007/s12630-010-9272-x.
74. Theiler L, Kleine-Brueggeney M, Urwyler N, Graf T, Luyet C, Greif R. Randomized clinical trial of the i-gel and Magill tracheal tube or single-use ILMA and ILMA tracheal tube for blind intubation in anesthetized patients with a predicted difficult airway. *Br J Anaesth.* 2011;107(2):243–50. doi:10.1093/bja/aer102.
75. Bhandari G, Shahi KS, Asad M, Parmar NK, Bhakuni R. To assess the efficacy of i-gel for ventilation, blind tracheal intubation and nasogastric tube insertion. *Anesth Essays Res.* 2013;7(1):94–9. doi:10.4103/0259-1162.114012.
76. Sastre JA, Lopez T, Garzon JC. [Blind tracheal intubation through two supraglottic devices: i-gel versus Fastrach intubating laryngeal mask airway (ILMA)]. *Rev Esp Anesthesiol Reanim.* 2012;59(2):71–6. doi:10.1016/j.redar.2012.02.016.
77. Driver BE, Scharber SK, Horton GB, Braude DA, Simpson NS, Reardon RF. Emergency department management of out-of-hospital laryngeal tubes. *Ann Emerg Med.* 2019;74(3):403–9. doi:10.1016/j.annemergmed.2019.01.025.
78. Burns JB, Jr., Branson R, Barnes SL, Tsuei BJ. Emergency airway placement by EMS providers: comparison between the King LT supralaryngeal airway and endotracheal intubation. *Prehosp Disaster Med.* 2010;25(1):92–5. doi:10.1017/S1049023X00007743.
79. Tumpach EA, Lutes M, Ford D, Lerner EB. The King LT versus the Combitube: flight crew performance and preference. *Prehosp Emerg Care.* 2009;13(3):324–8. doi:10.1080/10903120902935322.
80. Russi CS, Wilcox CL, House HR. The laryngeal tube device: a simple and timely adjunct to airway management. *Am J Emerg Med.* 2007;25(3):263–7. doi:10.1016/j.ajem.2006.03.018.
81. Bollig G, Lovhaug SW, Sagen O, Svendsen MV, Steen PA, Wik L. Airway management by paramedics using endotracheal intubation with a laryngoscope versus the oesophageal tracheal Combitube and EasyTube on manikins: a randomised experimental trial. *Resuscitation.* 2006;71(1):107–11. doi:10.1016/j.resuscitation.2006.02.016.
82. Abo BN, Hostler D, Wang HE. Does the type of out-of-hospital airway interfere with other cardiopulmonary resuscitation tasks? *Resuscitation.* 2007;72(2):234–9. doi:10.1016/j.resuscitation.2006.06.028.
83. Ritter SC, Guyette FX. Prehospital pediatric King LT-D use: a pilot study. *Prehosp Emerg Care.* 2011;15(3):401–4. doi:10.3109/10903127.2011.561400.
84. Saeedi M, Hajiseyedjavadi H, Seyedhosseini J, Eslami V, Sheikhmotaharvahedi H. Comparison of endotracheal intubation, combitube, and laryngeal mask airway between inexperienced and experienced emergency medical staff: A

- manikin study. *Int J Crit Illn Inj Sci.* 2014;4(4):303–8. doi:10.4103/2229-5151.147533.
85. Castle N, Owen R, Hann M, Naidoo R, Reeves D. Assessment of the speed and ease of insertion of three supraglottic airway devices by paramedics: a manikin study. *Emerg Med J.* 2010;27(11):860–3. doi:10.1136/emj.2009.084343.
 86. Leventis C, Chalkias A, Sampanis MA, Foulidou X, Xanthos T. Emergency airway management by paramedics: comparison between standard endotracheal intubation, laryngeal mask airway, and I-gel. *Eur J Emerg Med.* 2014;21(5):371–3. doi:10.1097/MEJ.0000000000000101.
 87. Jackson KM, Cook TM. Evaluation of four airway training manikins as patient simulators for the insertion of eight types of supraglottic airway devices. *Anaesthesia.* 2007;62(4):388–93. doi:10.1111/j.1365-2044.2007.04983.x.
 88. Genzwuerker HV, Fritz A, Hinkelbein J, Fintei T, Schlaefer A, Schaeffer M, Thil E, Rapp HJ. Prospective, randomized comparison of laryngeal tube and laryngeal mask airway in pediatric patients. *Paediatr Anaesth.* 2006;16(12):1251–6. doi:10.1111/j.1460-9592.2006.01984.x.
 89. Henlin T, Sotak M, Kovaricek P, Tyll T, Balcarek L, Michalek P. Comparison of five 2nd-generation supraglottic airway devices for airway management performed by novice military operators. *Biomed Res Int.* 2015;2015:1–8. doi:10.1155/2015/201898.
 90. Maddocks W. Skill fade in military medical training: A literature review of Supraglottic airway use in the prehospital environment. *Journal of Military and Veterans Health.* 2020;28(3):35–42.
 91. Fischer H, Hochbrugger E, Fast A, Hager H, Steinlechner B, Koinig H, Eisenburger P, Frantal S, Greif R. Performance of supraglottic airway devices and 12 month skill retention: a randomized controlled study with manikins. *Resuscitation.* 2011;82(3):326–31. doi:10.1016/j.resuscitation.2010.11.014.
 92. Ruetzler K, Roessler B, Potura L, Priemayr A, Robak O, Schuster E, Frass M. Performance and skill retention of intubation by paramedics using seven different airway devices—a manikin study. *Resuscitation.* 2011;82(5):593–7. doi:10.1016/j.resuscitation.2011.01.008.
 93. Bernhard M, Beres W, Timmermann A, Stepan R, Greim C-A, Kaisers UX, Gries A. Prehospital airway management using the laryngeal tube. An emergency department point of view. *Anaesthesist.* 2014;63(7):589–96. doi:10.1007/s00101-014-2348-1.
 94. Chinn M, Biedrzycki L. Prehospital laryngeal tube airway device placement resulting in hypopharyngeal perforation: a case report. *Prehosp Emerg Care.* 2020;24(4):590–4. doi:10.1080/10903127.2019.1671565.
 95. Vezina MC, Trepanier CA, Nicole PC, Lessard MR. Complications associated with the esophageal-tracheal combitube in the pre-hospital setting. *Can J Anaesth.* 2007;54(2):124–8. doi:10.1007/BF03022008.
 96. Vézina D, Lessard MR, Bussi  res J, Topping C, Tr  panier CA. Complications associated with the use of the esophageal-tracheal combitube. *Can J Anaesth.* 1998;45(1):76–80. doi:10.1007/BF03011999.
 97. Calkins TR, Miller K, Langdorf MI. Success and complication rates with prehospital placement of an esophageal-tracheal combitube as a rescue airway. *Prehosp Disaster Med.* 2006;21(2):97–100. doi:10.1017/s1049023x00003423.
 98. Oczenski W, Krenn H, Dahaba AA, Binder M, El-Schahawi-Kienzl I, Kohout S, Schwarz S, Fitzgerald RD. Complications following the use of the Combitube, tracheal tube and laryngeal mask airway. *Anaesthesia.* 1999;54(12):1161–5. doi:10.1046/j.1365-2044.1999.01107.x.
 99. Segal N, Yannopoulos D, Mahoney BD, Frascione RJ, Matsuura T, Cowles CG, McKnite SH, Chase DG. Impairment of carotid artery blood flow by supraglottic airway use in a swine model of cardiac arrest. *Resuscitation.* 2012;83(8):1025–30. doi:10.1016/j.resuscitation.2012.03.025.
 100. Kim TH, Hong KJ, Shin SD, Lee JC, Choi DS, Chang I, Joo YH, Ro YS, Song KJ. Effect of endotracheal intubation and supraglottic airway device placement during cardiopulmonary resuscitation on carotid blood flow over resuscitation time: An experimental porcine cardiac arrest study. *Resuscitation.* 2019;139:269–74. doi:10.1016/j.resuscitation.2019.04.020.
 101. White JMB, Braude DA, Lorenzo G, Hart BL. Radiographic evaluation of carotid artery compression in patients with extraglottic airway devices in place. *Acad Emerg Med.* 2015;22(5):636–8. doi:10.1111/acem.12647.
 102. Neill A, Ducanto J, Amoli S. Anatomical relationships of the Air-Q supraglottic airway during elective MRI scan of brain and neck. *Resuscitation.* 2012;83(12):e231–e232. doi:10.1016/j.resuscitation.2012.08.322.
 103. Nandwani N, Fairfield MC, Krarup K, Thompson J. The effect of laryngeal mask airway insertion on the position of the internal jugular vein. *Anaesthesia.* 1997;52(1):77–9. doi:10.1111/j.1365-2044.1997.012-az012.x.
 104. Rasulo F, Zugni N, Piva S, Fagoni N, Pe F, Toninelli A, et al. Influence of supraglottic airway device placement on cerebral hemodynamics. *Minerva Anestesiol.* 2016;82(8):850–7.
 105. Eismann H, Sieg L, Otten O, Leffler A, Palmaers T. Impact of the laryngeal tube as supraglottic airway device on blood flow of the internal carotid artery in patients undergoing general anaesthesia. *Resuscitation.* 2019;138:141–5. doi:10.1016/j.resuscitation.2019.03.010.
 106. Colbert S-A, Ohanlon DM, Flanagan F, Page R, Moriarty DC. The laryngeal mask airway reduces blood flow in the common carotid artery bulb. *Can J Anaesth.* 1998;45(1):23–7. doi:10.1007/BF03011987.
 107. Zhang J-J, Qu Z-Y, Hua Z, Zuo M-Z, Zhang H-Y. Effect of different types of laryngeal mask airway placement on the right internal jugular vein: A prospective randomized controlled trial. *WJCC.* 2019;7(24):4245–53. doi:10.12998/wjcc.v7.i24.4245.
 108. Ruetzler K, Gruber C, Nabecker S, Wohlfarth P, Priemayr A, Frass M, Kimberger O, Sessler DI, Roessler B. Hands-off time during insertion of six airway devices during cardiopulmonary resuscitation: a randomised manikin trial. *Resuscitation.* 2011;82(8):1060–3. doi:10.1016/j.resuscitation.2011.03.027.
 109. Wang HE, Schmicker RH, Daya MR, Stephens SW, Idris AH, Carlson JN, Colella MR, Herren H, Hansen M, Richmond NJ, et al. Effect of a strategy of initial laryngeal tube insertion vs endotracheal intubation on 72-hour survival in adults with out-of-hospital cardiac arrest: a randomized clinical trial. *JAMA.* 2018;320(8):769–78. doi:10.1001/jama.2018.7044.
 110. Bengert JR, Kirby K, Black S, Brett SJ, Clout M, Lazaroo MJ, Nolan JP, Reeves BC, Robinson M, Scott LJ, et al. Effect of a strategy of a supraglottic airway device vs tracheal intubation during out-of-hospital cardiac arrest on functional outcome: the AIRWAYS-2 randomized clinical trial. *JAMA.* 2018;320(8):779–91. doi:10.1001/jama.2018.11597.
 111. Hardy G, Maddry JK, Ng PC, Savell SC, Arana AA, Kester A, Beberta VS. Impact of prehospital airway management on combat mortality. *Am J Emerg Med.* 2019;37(2):349–50. doi:10.1016/j.ajem.2018.02.007.
 112. Hernandez MC, Antiel RM, Balakrishnan K, Zielinski MD, Klinkner DB. Definitive airway management after prehospital supraglottic rescue airway in pediatric trauma. *Journal of Pediatric Surgery.* 2018;53(2):352–6. doi:10.1016/j.jpedsurg.2017.10.004.